

Synthesis and electrochemical properties of $\text{Li}_2\text{S}-\text{B}_2\text{S}_3-\text{Li}_4\text{SiO}_4$

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Abstract

New Li^+ ion-conductive glasses $\text{Li}_2\text{S}-\text{B}_2\text{S}_3-\text{Li}_4\text{SiO}_4$ were synthesized by rapid quenching, and they were transformed into glass ceramics by heat treatment. The heat treatment increased the ionic conductivities of the Li_4SiO_4 -doped glasses, and the highest ionic conductivity observed in the system was $1.0 \times 10^{-3} \text{ S cm}^{-1}$ at room temperature. The glass ceramics were highly stable against electrochemical oxidation with a wide electrochemical window of 10 V.

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1. Introduction

Li^+ ion batteries are a key component of the portable, entertainment, computing, and telecommunication equipment. Such batteries, however, have involved the issue of safety [1] due to flammable organic solvents used in the electrolytes. Solid electrolytes are very attractive for application to lithium batteries, because they will solve the safety problem of lithium batteries.

Most of the studies on solid electrolytes have been mainly focused on their ionic conductivity. However, solid electrolytes for lithium batteries are required to have not only high lithium ionic conductivity but also electrochemical stability such as wide electrochemical window, because they should be stable to highly oxidative cathodes and highly reactive anodes. Ionic conductivities of the order of $10^{-4} \text{ S cm}^{-1}$ in sulfide-based glasses including $\text{Li}_2\text{S}-\text{SiS}_2$ [2–8], $\text{Li}_2\text{S}-\text{P}_2\text{S}_5$ [9–12] and $\text{Li}_2\text{S}-\text{B}_2\text{S}_3$ [13] have been enhanced to $10^{-3} \text{ S cm}^{-1}$ by doping LiI . How-

ever, the doping makes them unstable against electrochemical oxidation, because their I^- ions are oxidized at 2.9 V vs. Li/Li^+ [9]. Recently, the system of $\text{SiS}_2-\text{Li}_2\text{S}$ glasses doped with *ortho*-oxosalts such as Li_3PO_4 [14] and Li_4SiO_4 [6] were reported to have high ionic conductivities of the order of $10^{-3} \text{ S cm}^{-1}$ without LiI -doping. The glasses have an electrochemical window wider than 10 V owing to the absence of I^- ions.

The present study is focused on whether or not the similar doping of lithium *ortho*-oxysalts enhances the ionic conductivity of $\text{B}_2\text{S}_3-\text{Li}_2\text{S}$. Lithium *ortho*-silicate (Li_4SiO_4) was used as an *ortho*-oxysalt. In addition, the $\text{Li}_2\text{S}-\text{B}_2\text{S}_3-\text{Li}_4\text{SiO}_4$ glasses were crystallized into glass ceramics, because some of the crystalline phases obtained from some ion-conductive glasses by heat treatment have been reported to be more conductive than the glasses [10–12]. The ion conductivities and electrochemical stability are reported.

2. Experimental

The composition of the base $\text{Li}_2\text{S}-\text{B}_2\text{S}_3$ glass was chosen to be 0.7 $\text{Li}_2\text{S}-0.3 \text{ B}_2\text{S}_3$, which had already been reported to show the highest conductivity in the system [15]. First of all, B_2S_3 was synthesized from amorphous boron (B, 99.9%, Stream Chemical) and sulfur (S, 99.9%, Rare Metallic) by the

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procedure described in the system [16]. The resultant B_2S_3 was mixed with lithium sulfide (Li_2S , 99.9%, Idemitsu) and lithium ortho-silicate (Li_4SiO_4 , 99.9%, Aldrich), and the mixture was sealed in a carbon coated quartz tube and heated at 800 °C for 2 h in an electric furnace. The molten sample was rapidly quenched in ice-water (herein after referred to as-quenched sample). The glass ceramics were synthesized from the glass by heat treatment just above their crystallization temperatures.

The obtained samples were characterized by powder X-ray diffraction (XRD) and differential scanning calorimetry (DSC). The XRD data were collected on a diffractometer (Rigaku, RINT-2000) using Cu *K* α radiation. DCS was carried out for a powdered sample sealed in a Pyrex tube by using a scanning calorimeter (DSC 111, Setaram) with a heating rate of 3 °C min⁻¹.

Electric conductivities of the samples were measured for cold-pressed powder under an Ar gas flow in the temperature range between 25 °C and 200 °C by an a.c. impedance method using an impedance analyzer (Solartron 1260). Complex impedance data were taken in a frequency range from 100 Hz to 10 MHz. The graphite powder was used as electrodes; which was pressed together with an electrolyte layer on its both sides. The electrochemical stability of the obtained samples was investigated by cyclic voltammetry. An electrochemical cell of the solid electrolyte was constructed as follows.

The ground solid electrolyte powder was put in a hole with a diameter of 10 mm of insulator tube and pressed into a pellet. A stainless steel (SUS 304) plate with an electrode area of 0.785 cm² and a lithium foil were attached onto the faces of the pellet as working and counter electrodes, respectively. The cyclic voltammogram was obtained using a potentio–galvano stat (Solortron 1286) at a scanning rate of 5 mV s⁻¹ and in the range between -0.5 and 10 V.

3. Results and discussion

The material obtained by quenching the molten of the Li_2S – B_2S_3 – Li_4SiO_4 mixture did not give any reflections in its XRD

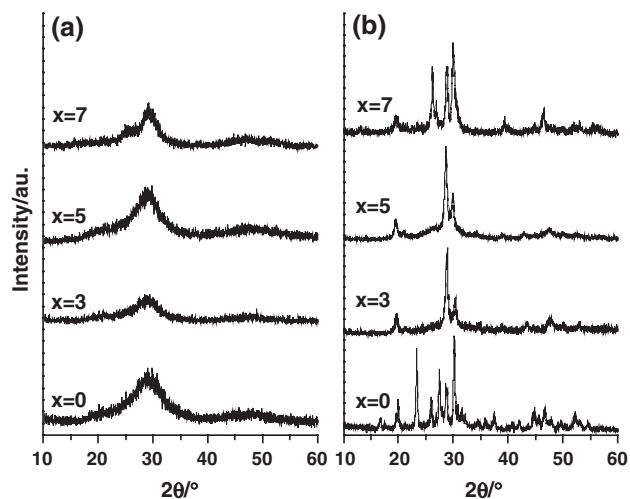


Fig. 1. Powder XRD patterns of (a) as-quenched and (b) after heat treatment samples of $(1-x)(0.7Li_2S-0.3B_2S_3)-xLi_4SiO_4$.

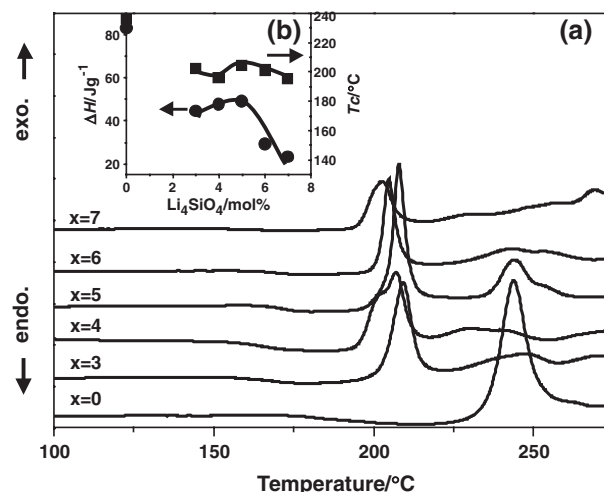


Fig. 2. (a) DSC curves of as-quenched sample and (b) composition dependence of the enthalpy (ΔH) and crystallization temperature (T_c) of $(1-x)(0.7Li_2S-0.3B_2S_3)-xLi_4SiO_4$ glass.

pattern; halos were observed instead as shown in Fig. 1(a). In the DSC measurement, they showed glass transitions and sharp exothermic peaks corresponding to their crystallization in the temperature ranges from 163 to 173 °C and from 195 to 204 °C, respectively (Fig. 2(a)). These results evidenced the glassy nature of the as-quenched materials. Fig. 3 shows the composition dependence of electrical conductivity (σ) and activation energy for conduction (E_a) of the $(1-x)(0.7Li_2S-0.3B_2S_3)-xLi_4SiO_4$ glasses and glass ceramics. Addition of Li_4SiO_4 slightly increased the ionic conductivity ($\sigma_{25\text{ °C}}$) of the $0.7Li_2S-0.3B_2S_3$ glass; $\sigma_{25\text{ °C}}$ for the $0.7Li_2S-0.3B_2S_3$ glass (2.1×10^{-4} S cm⁻¹) was increased to 3.5×10^{-4} S cm⁻¹ by 5 mol% doping of Li_4SiO_4 .

The heat treatment decreased the $\sigma_{25\text{ °C}}$ of the non-doped glasses, while it increased those of the doped ones. It suggests that the crystalline phase formed from the doped glass was different from that from the non-doped glass; the former was highly conductive, while the latter poorly conductive. In fact, crystallization temperatures for the doped glasses were quite

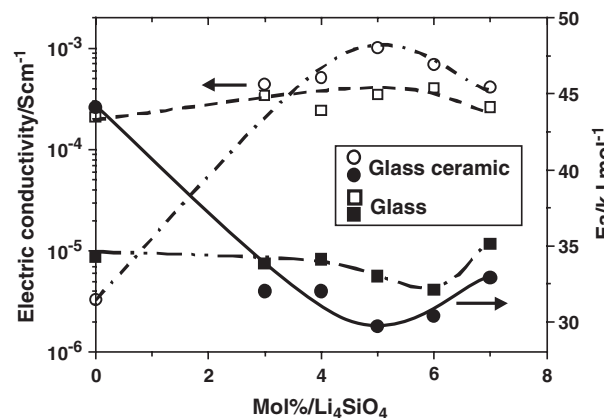


Fig. 3. Composition dependence of the electrical conductivities at 25 °C (σ_{25}) and activation energies for conduction (E_a) of the $(1-x)(0.7Li_2S-0.3B_2S_3)-xLi_4SiO_4$.

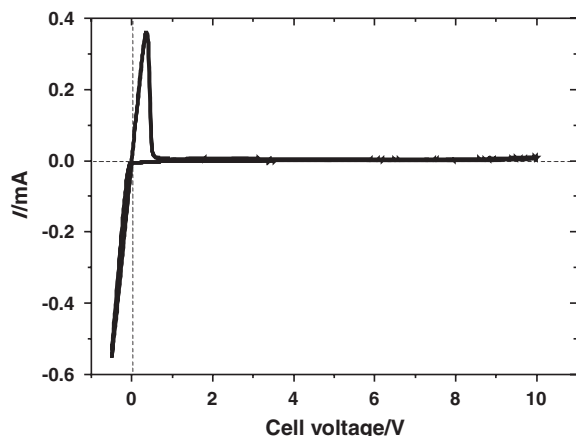


Fig. 4. Cyclic voltammogram of the 0.95(0.7Li₂S–0.3B₂S₃)–0.05Li₄SiO₄ glass. The voltammogram was performed at room temperature with a scanning rate 5 mV s^{−1} and the cell area is 0.785 cm².

different from that for the non-doped one, and the XRD patterns for the former ($x=3$ and 5 mol%) and the latter were quite different. For the glass ceramics after heat treatment, the $\sigma_{25\text{ }^{\circ}\text{C}}$ increased with increasing x and reached a maximum ($1.0 \times 10^{-3} \text{ S cm}^{-1}$) at $x=5$ mol%, at which E_a showed a minimum E_a of 29.7 kJ mol^{−1}. If the Li₄SiO₄-doped glasses were crystallized into more highly conductive glass ceramics as discussed above, the enthalpy (ΔH) in the crystallization peak should be proportional to the fraction of the highly conductive crystalline phase in the glass ceramics after the heat treatment and thus show a similar change to the $\sigma_{25\text{ }^{\circ}\text{C}}$. In fact, change in $\sigma_{25\text{ }^{\circ}\text{C}}$ was in coincidence the change in ΔH ; ΔH showed a maximum at $x=5$ mol% as shown in Fig. 2(b). Further addition of Li₄SiO₄ up to 7 mol% brought about new reflections in the XRD pattern. Taking into account that the $\sigma_{25\text{ }^{\circ}\text{C}}$ decreased in the composition range, the crystalline phase giving the new reflections should not be as highly conductive as that discussed above.

Fig. 4 shows the cyclic voltammogram for the 95(0.7Li₂S–0.3B₂S₃)–5Li₄SiO₄ glass. The lithium deposition (Li⁺+e[−]→Li) and dissolution (Li→Li⁺+e[−]) reactions were observed in the potential range from −0.5 to 0.3 V. No other reactions were

observed in the potential range up to 10 V vs. Li/Li⁺, indicating a wide electrochemical window of the 0.95(0.7Li₂S–0.3B₂S₃)–0.05Li₄SiO₄. The above results indicate that doping of lithium *ortho*-silicate Li₄SiO₄ improved the ionic conductivity of B₂S₃–Li₂S to be 10^{−3} S cm^{−1} without narrowing the electrochemical window.

4. Conclusion

We synthesized Li₄SiO₄-doped Li₂S–B₂S₃ glasses by the rapid-quenching and transformed them into glass ceramics by heat treatment. When 5 mol% of Li₄SiO₄ was doped to the 0.7Li₂S–0.3B₂S₃, the glass ceramic showed an ionic conductivity of $1 \times 10^{-3} \text{ S cm}^{-1}$ and wide electrochemical window of about 10 V.

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